



Prathmesh Ojha

B.Tech CSE (Data Science and Artificial Intelligence) — 2027
8595608059 — prathmesh.ojha.23cse@bmu.edu.in — LinkedIn — GitHub
Portfolio — Engineering Playbook — Blueprint

Internship

Getting Roots (*Mobile Application Developer*)

Jun 2025 - Jul 2025

- Developed a cross-platform Flutter application (iOS, Android) following Clean Architecture principles, implementing modular features with Provider-based state management.
- Integrated REST APIs with remote/local data flows, local caching fallback, dependency injection (GetIt), and structured error handling.

Projects

Angel Five (AI Financial Dashboard)

Sep 2025 – Present — Demo

Tech: Next.js, React, TypeScript, Node.js, Express, FastAPI, PostgreSQL, Firebase, Redis, BullMQ

- Built a full-stack financial dashboard using **React/Next.js** and **Node.js/Express**, implementing watchlists, portfolio analytics, authentication, notifications, and market-data workflows.
- Designed and integrated **RESTful APIs** across frontend, backend, and ML services, handling authentication, validation, asynchronous operations, and service communication.
- Implemented **Redis/BullMQ** for asynchronous market-data ingestion, forecasting, analytics, and notification workloads to improve application responsiveness.
- Developed modular backend services with **Express middleware, async/await, error handling, structured logging, Docker, and Render/Vercel deployment.**

Rapid AI Drug Repurposing (Research Project)

Feb 2026 – Present

Tech: Python, PyTorch Geometric, PyTorch, PrimeKG, Ollama (LLaMA 3.2), Streamlit

- Built an inductive **GraphSAGE** framework on PrimeKG (10.6K nodes, 161.6K edges) to rank candidates across 2.45M drug-disease pairs.
- Engineered 131D multimodal features from TF-IDF biomedical embeddings and physicochemical molecular descriptors.
- Achieved **92.35% AUC-ROC** and **84.84% F1** ($92.63\% \pm 0.41\%$ 5-seed CV) with strict leakage prevention and GCN/GAT/RGCN benchmarking.
- Deployed a Streamlit Discovery Lab with < 2.5 ms CPU inference and local LLaMA 3.2 mechanistic explanations.

Skills

Programming Languages: Python, JavaScript, TypeScript, C++, Java, SQL, Dart

Web & App Development: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Flutter, Tailwind CSS

Databases & Tools: PostgreSQL, MySQL, MongoDB, Firebase, Redis, Docker, Render, Vercel, Git, GitHub, BullMQ

Concepts: REST APIs, API Design, Async/Await, Middleware, Microservices, CI/CD, Data Structures & Algorithms, OOP, Database Design, Authentication, State Management, Debugging, Problem Solving

AI & Machine Learning: PyTorch, Hugging Face, Scikit-Learn, Pandas, NumPy

Education

B.Tech CSE (Data Science and Artificial Intelligence)

2023 – 2027

CGPA: 7.57

Credit-Linked Program in Data Science, Daksh Gurukul, IIT Guwahati

2025 – 2026